AMRITSAR GROUP OF COLLEGES NAAC Grade "A" 3rd Cycle | Autonomous College under Autonomous Category | (Since 2014) Conferred by UGC	AGCS-21704 Major Project Lab	Department of Computer Science and Engineering

B.Tech (CSE) 7th Sem – 3 (Group G1)

FORTNIGHTLY PROJECT PROGRESS REPORT-6

Date : 01-10-2026 to 15-10-2026
Group number : 07

Project Title : Nexora – An Intelligent Knowledge Workspace

Project Members:

S.No	Univ. Roll No	Name	Role
1	2333428	Shivam Kumar	Member
2	2333429	Shivam Kumar Pandit	Member
3	2333436	Sonu Kamat	Syndicate Leader

Progress Summary	1. Developed multi-stage Dockerfile optimizing container size (< 450 MB footprint) for backend API and dependencies. 2. Created docker-compose.yml configuration to launch FastAPI backend, ChromaDB persistent volumes, and frontend services. 3. Configured GitHub Actions CI/CD automated pipeline (.github/workflows/ci.yml) for linting (flake8), type checking (mypy), and unit tests. 4. Conducted comprehensive concurrency load testing using Locust and pytest-asyncio simulating 100 simultaneous active users. 5. Verified 99.9% system uptime, zero memory leaks, and CPU load under 35% during peak load testing. 6. Finalized 12-slide Major Project PowerPoint Presentation (Nexora_12_Slide_Presentation.pptx). 7. Prepared complete Project Synopsis, System Operating Manual, and Viva Voce Q&A; Reference Handbook. 8. Conducted end-to-end integration testing across all system modules: Document Processing, Vector Engine, Multi-Agent Swarm, Hybrid Search, Auth, and Voice UI. 9. Deployed and verified live production-ready instance at http://127.0.0.1:8000. 10. Completed full project codebase sync and pushed updates to GitHub repository (https://github.com/Sonukamat/Nexora-workspace).
Work In Progress	1. Final rehearsal for the Major Project Viva Voce defense presentation. 2. Submission of final project report, progress reports, and demonstration video to the department evaluation committee. 3. Preparing project repository release tags and final documentation archive.

Challenges And Issues	1. Reducing Docker image size while retaining heavy machine learning libraries (torch, SentenceTransformers, ChromaDB). 2. Managing resource allocation and memory usage during multi-user concurrent query processing in Docker containers. 3. Ensuring seamless cross-platform deployment across Windows, Linux, and macOS environments. 4. Resolving asynchronous database connection bottlenecks under high-concurrency load tests. 5. Maintaining backward compatibility of API endpoints during final deployment build steps.